

**MTH302**

**Final Term (Live Quiz)**

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**Past Papers for Mids./Finals are also Available**

**Subject Code: MTH302 - Business Mathematics & Statistics**

**Question #1:**

If  $S_{xy} = -3.27$ ,  $S_x = 2.416$  and  $r = -0.091$ , then Standard deviation of Y is

- 12.85
- 15.88 ✓
- 14.87
- 13.86

**Question #2:**

The intercept of a line passing through origin is

- finite
- positive
- zero ✓
- does not exists

**Question #3:**

If  $s_x=2$ ,  $s_y = 4$  and  $s_{xy} =10$  then correlation coefficient 'r' is

- 1.12
- 0.80
- 1.25 ✓
- 3.54

**Question #4:**

In the regression line equation  $Y=a+bX$ ; b is called the \_\_\_\_\_.

- Slope ✓

- X-intercept
- Y-intercept
- None of these

**Question #5:**

If  $S_{xy} = -5.5$ ,  $S_y = 2.35$  and  $r = -0.08$ , then Standard deviation of X is

- 15.77
- 20.47 ✓
- 29.25
- 12.85

**Question #6:**

What does the term "outlier" mean in the context of a scatter plot with regression analysis?

- A data point that deviates significantly from the overall pattern ✓
- A data point that lies close to the regression line
- The slope of the regression line
- The intercept of the regression line

**Question #7:**

If the correlation is negative perfect, what will be the value of  $r$ ?

- 1
- -1 ✓
- 0
- None of these

**Question #8:**

If scores of openness to experience tend to be higher for students who have higher interest for adventure, which of the following is a reasonable correlation between openness to experience and interest for adventure?

- -0.78
- +0.21
- +0.78 ✓
- -0.21

**Question #9:**

The regression model explains the \_\_\_\_\_ between two or more variables in the given population.

- Association ✓
- Connection
- All of these
- Relation

**Question #10:**

Intercept of the regression line can be defined using the equation  $a=Y-bX$ .

- False
- True ✓

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**Question # 2**

If the estimating equation is  $Y = a - b X$ , which of the following is true?

- The y intercept is b
  - There is inverse relationship ✓
  - Slope of line is positive
  - all of these
- 

**Question # 3**

The predicted rate of response of the dependent variable to changes in the independent variable is called .....

- Slope ✓
  - Error
  - Intercept
  - Coefficient of determination
- 

**Question # 4**

Suppose that samples of polythene bags from two manufacturers A and B are tested by a buyer for bursting pressure. Following results on the basis of sample have been observed:

Manufacturer A: Mean = 21, Standard Deviation= 4.875 and Coefficient of Variation = 23.32%

Manufacturer B: Mean = 21.81, Standard Deviation = 7.074 and Coefficient of Variation = 32.44%

Which manufacturer's bag has more uniform pressure?

- Manufacturer B
  - Manufacturer A ☒
- 

**Question # 5**

Which statistical test is commonly used to assess the significance of a correlation coefficient?

- Pearson correlation test
  - Chi-square test
  - None of these
  - t-test ☒
- 

**Question # 6**

In regression equation  $Y = a + bX$ ,

- b represents the slope of the line. ☒
  - b represents the dependent variable.
  - b represents the value of the intercept.
  - b represents the independent variable.
- 

**Question # 7**

If scores of openness to experience tend to be higher for students who have higher interest for adventure, which of the following is a reasonable correlation between openness to experience and interest for adventure?

- +0.21
  - -0.21
  - -0.78
  - +0.78 ☒
- 

**Question # 8**

Seasonal variation is defined as \_\_\_\_\_.

- Actual – trend ☒
- Actual × trend
- None of these

- Actual + trend
- 

### Question # 9

Equation of line having slope 0 and passing through the point A (0, 0) is

- $Y = X$
  - $Y = 0$  ☒
  - $X = 0$
  - $X - 1 = 0$
- 

### Question # 10

The moving averages of the Prices 20, 30, 40, 50, 60 are

- 30, 40, 50
- 20, 40, 60
- 40, 40, 40 ☒
- 35, 45, 55

Fashion styles buy shirts for Rs. 500 and sell them for Rs. 1200. Find the rate of markup based on the cost?

- Rs. 714%
- Rs. 140% ☒
- Rs. 40%
- Rs. 417%

? What is the relationship between selling price, cost and markup?

- $\text{Cost} = \text{Selling price} + \text{Mark up}$
- $\text{Selling price} = \text{Cost} - \text{Mark up}$
- $\text{Mark up} = \text{Selling price} + \text{Cost}$
- $\text{Selling price} = \text{Cost} + \text{Mark up}$  ☒

? The correct relation among %markup on sale, cost price and selling price is...

- $\text{Selling Price} = \text{Cost price} + (\text{Selling price} \times \% \text{Markup on sale})$
- None of these

- Selling Price = Cost price + (cost price x %Markup on sale) ✓
- Cost Price = Cost price + (Selling price x %Markup on sale)

❓ If L is the current price of an item and a series discount of 10%, 15% and 13% are given. What is the new price of the item?

- $L (1-10\%) (1-15\%) (1-13\%)$  ✓
- $L (1-10\% - 15\% - 13\%)$
- $L (1 - 38\%)$
- $L (1+10\%) (1+15\%) (1+13\%)$

❓ A book sells for Rs. 50. It has just been reduced 40%. What will be the selling price?

- Rs. 20
- Rs. 10
- Rs. 30 ✓
- Rs. 40

❓ Which of the following function returns the depreciation of an asset for a specified period?

- MIRR
- IRR
- DB
- DDB ✓

❓ ATV whose catalogue price is Rs. 15000 is available at two successive discounts of 10% and 20%. What is the single equivalent discount?

- 25%
- 28% ✓
- 32%

❓ The original sale price can be estimated as bought price multiplicative factor of ratio:

- Assumed selling price and markup ✓
- Assumed selling price and cost
- None of these
- Assumed selling price and markdown

❑ If some percentage is given as markup, without mentioning that whether it is markup on cost or markup on sale, it is evident that %markup on ---- is under consideration.

- Sale
- Cost ☒

❑ An item originally priced at Rs. 60 is marked 20% off. What is the sale price?

- Rs. 48 ☒
- Rs. 45
- Rs. 51

1. An automatic production line breaks down every 2 hours.

- Options:

- a. 0.312
- b. None of these
- c. 0.018
- d. 0.021

- Correct Answer: c. 0.018

2. The successive negative binomial trials are assumed to be:

- Options:

- a. Dependent
- b. May be independent
- c. None of these
- d. Independent

- Correct Answer: a. Dependent

3. The probability of a sure event is:

- Options:

- a. None of these
- b. 0

- c. 1
- d. -1

- Correct Answer: c. 1

4. The Excel command "`=BINOMDIST( -2, 5, 0.5, FALSE)`" gives an error of #NUM. The reason is that the last parameter is:

- Options:

- a. FALSE
- b. True
- c. None of these
- d. Independent

- Correct Answer: a. FALSE

5. There are 5 white balls and 4 black balls. In how many ways can we select 3 white and 2 black balls?

- Options:

- a. 50
- b. 60
- c. 40
- d. 30

- Correct Answer: b. 60

6. The number of combinations of 3 different objects A, B, C taken two at a time is:

- Options:

- a. 3
- b. 4
- c. 5
- d. 6

- Correct Answer: a. 3

7. When "CRITBINOM" function will return "#NUM" error?

- Options:

- a. When  $\alpha = 1$
- b. None of these
- c. None of these
- d. When  $\alpha > 1$

- Correct Answer: d. When  $\alpha > 1$

8. In Binomial distribution, the outcome on one trial:

- Options:

- a. May affect
- b. Does affect
- c. Does not affect
- d. None of these

- Correct Answer: c. Does not affect

9. The probability of successes can be less than 0 or greater than 1.

- Options:

- a. True
- b. False
- c. None of these
- d. Independent

- Correct Answer: b. False

10. What is the value of  ${}^5C_5$  (C for combination)?

- Options:

- a. 0
- b. 1
- c. 3

d. 5

- Correct Answer: b. 1

1. Which of the following is true for the 1! (one factorial)?

Options:

A) 1

B) 2

C) 3

D) 4

Correct Option: A) 1

2. In the context of the PERMUT function, what does the second argument represent?

Options:

A) The number of items to be chosen.

B) The factorial of a number.

C) The square root of a number.

D) The total number of items.

Correct Option: A) The number of items to be chosen.

3. In Excel's NEGBINOMDIST function, what does the argument "number\_s" represent?

Options:

A) The threshold number of successes

B) The cumulative probability

C) The number of trials

D) The probability of success

Correct Option: C) The number of trials

4. If the probability  $P(\text{you pass}) = 1$ , then the probability  $P(\text{you fail}) = \dots$

Options:

- A) 0.5
- B) 0.7
- C) 0
- D) 1

Correct Option: C) 0

5. In general  $n!$  = \_\_\_\_\_

Options:

- A)  $n$  factorial ( $n!$ )
- B) None of these
- C) Many number of times
- D) Fixed number of times

Correct Option: A)  $n$  factorial ( $n!$ )

6. When A and B are mutually exclusive, then  $P(A \text{ or } B) = \dots$

Options:

- A)  $P(A) - P(B)$
- B)  $P(A) + P(B)$
- C)  $P(A) * P(B)$
- D)  $P(A) / P(B)$

Correct Option: B)  $P(A) + P(B)$

7. A fair coin is tossed three times then its total number of sample points of the sample space will be ...

Options:

- A)  $2^4$
- B)  $3^2$
- C)  $2^3$
- D) None of these

Correct Option: C)  $2^3$

8. What does the CRITBINOM function in Excel calculate?

Options:

- A) Returns the smallest value for which cumulative binomial distribution  $\leq$  criterion value.
- B) Returns the smallest value for which cumulative binomial distribution  $\geq$  criterion value.
- C) Returns the greatest value for which cumulative binomial distribution  $\leq$  criterion value.
- D) Returns the greatest value for which cumulative binomial distribution  $\geq$  criterion value.

Correct Option: B) Returns the smallest value for which cumulative binomial distribution  $\geq$  criterion value.

9. In binomial distribution, the experiment is repeated for ...

Options:

- A) None of these
- B) Many number of times
- C) Fixed number of times
- D) Variable number of times

Correct Option: C) Fixed number of times

10. Poisson probability density function can never assume the value ...

Options:

- A) 0
- B) 0.5
- C) -0.5
- D) None of these

Correct Option: D) None of these

1. The following formula for probability:  $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$  is called:

- A) None of these
- B) The law of subtraction
- C) The law of addition

- D) The law of multiplication
- Correct Option: C) The law of addition

2. Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.

- A) False
- B) True
- Correct Option: B) True

3. Which statement about the negative binomial distribution is correct?

- A) Each trial has just two possible outcomes
- B) Each trial has just two possible outcomes
- C) The experiment consists of
- D) Its trials are independent
- Correct Option: A) Each trial has just two possible outcomes

4. In how many ways can a cricket eleven be chosen out of a batch of seventeen players?

- A)  ${}^{11}C_{17}$
- B)  ${}^{17}C_{11}$
- C)  ${}^{17}C_4$
- D)  ${}^{11}C_{11}$
- Correct Option: B)  ${}^{17}C_{11}$

5. In the context of the PERMUT function, what does the first argument represent?

- A) The factorial of a number.
- B) The total number of items.
- C) The square root of a number.
- Correct Option: B) The total number of items.

6. In how many ways can the letters of the word 'APPLE' be arranged?

- A) 60
- B) 720
- C) 120
- D) 180
- Correct Option: A) 60

7. The mathematical formula for the Exponential Smoothing is:

- A) New Forecast = Old Forecast - (Actual – old Forecast)
- B) New Forecast = old Forecast + (Actual – old Forecast)
- C) New Forecast = Old Forecast - (Actual + old Forecast)
- D) New Forecast = old Forecast + (Actual + old Forecast)
- Correct Option: B) New Forecast = old Forecast + (Actual – old Forecast)

8. A fair coin is tossed three times then its total number of sample points of the sample space will be:

- A) None of these
- B)  $3^2$
- C)  $3^4$
- Correct Option: B)  $3^2 = 9$

9. The trials of the negative binomial distribution are independent.

- A) False
- B) True
- Correct Option: B) True

10. In the context of a dialog box, the output range must be on the \_\_\_\_ worksheet as the data used in the input range.

- A) Same
- B) Previous

- C) Adjacent
- D) Next
- Correct Option: A) Same

math302

Question 1: If the equation of regression line is  $y = 6$ , then which of the following will be true for it.

The line is parallel to x-axis ☒

The line passes through the origin

Slope of the line is equal to 6

The line is parallel to y-axis

Question 2: If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason for the error?

Either Array1 or array2 is empty. ☒

Array or reference argument contains text, logical values, or empty cells.

The arguments are names, arrays, or references that contain numbers.

Array1 and Array2 have different numbers of data points.

Question 3: The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as:

the standard deviation

Coefficient of variation ☒

the expected return

a probability distribution

Question 4: In the model  $y = mx + a$ ,  $y$  is also known as the .....

Predictor variable

Predicted (dependent) variable ☒

Independent variable

Explanatory variable

Question 5: If many data points are close to the mean, then the standard deviation is -----.

Middle value

Large

0

Small ☒

Question 6: The value of the coefficient of variation (CV) is:

Same unit as data

In percentage

Unitless

Both unitless and in percentage ☒

Question 7: If Mean of X = 7.857, Mean of Y = 4.857, and Slope = -0.47538, then intercept of the data is:

5.627 ☒

7.911

9.978

8.592

Question 8: For the least squares regression line  $y = 0.9 + 1.7x$ , \_\_\_\_ and \_\_\_\_ are slope and intercept respectively.

1.7 and 0.9 ☒

0 and 1

2.6 and 0

0.9 and 1.7

Question 9: For the data set below:

Copy code

X Y

3 6

5 8

7 11

9 14

11 16

=INTERCEPT(D7:D11,C7:C11) = 1.9

The Intercept function gives the value of y-intercept, which is 1.9 in the above question. Keeping in view the definition of intercept, what will be the value of x when y = 1.9?

x = 2 ☒

x = 1.3

x = 1

x = 0

Question 10: For small to moderate sample size, the sample standard deviation is \_\_\_\_\_.

Underestimated ☒

Overestimated

Unbiased

None of these

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Question # 1:

A graphical method of representing the sample points of a multiple-step experiment is

A) Tree diagram ☒

B) An ogive

C) A histogram

D) A frequency polygon

Question # 2:

A type of graph that shows change over time is a

- A) Circle graph
- B) Double-bar graph
- C) Bar graph
- D) Line graph ☒

Question # 3:

The average of the first and third quartiles is called...

- A) Midhinge ☒
- B) Midrange
- C) Interquartile range
- D) Range

Question # 4:

.....use the division of a circle into different sectors.

- A) Line graph
- B) Frequency Polygon
- C) Sector graphs ☒
- D) Conversion Graphs

Question # 5:

.....is used in compounding such as investments that are accumulated over a period of time

- A) Geometric mean ☒
- B) Arithmetic mean
- C) Harmonic mean
- D) Mid range

Question # 6:

If some values are given then their mean may be .....

- A) Zero
- B) Positive
- C) All above ☒
- D) Negative

Question # 7:

The median of ungroup data is

- A) Both above
- B) Middle value, if values are odd. ☒
- C) Middle value, if values are even
- D) None of these

Question # 8:

Which of the following is not a type of graph?

- A) Frequency distribution ☒
- B) Circle graph
- C) Column graph
- D) None of these

Question # 9:

The statement that the relative frequency can be converted into cumulative frequency by adding the current frequency to the previous total is

- A) False
- B) True ☒

Question # 10:

The mode of the data 6, 6, 9, 9, 9, 10, 11, 11, 15, 18 is

A) 11

B) 6

C) There is no mode for this data

D) 9 ☒

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Question 1:

The range of the data 2, 5, 8, 9, 15, 6, 7 is ....

10

9

11

13 ☒

Question 2:

The graph below shows the rainfall over a period of 12 months. Find the month in which there was the second least rainfall.

June ☒

January

August

November

Question 3:

Relative frequency is defined as

None of these

the ratio of frequency of the class and total frequency ☒

the ratio of frequency of the class to the highest frequency

frequency obtained by adding the previous to the next class frequency

Question 4:

If the value '2' is collected in the data  $n/2$  times and '3' occurs  $n/2$  times, then the arithmetic mean is

5/2 ✓

None of these

5n/2

5(n-1)/2

Question 5:

The average of smallest and largest values in the data is called.....

Range

Midrange ✓

Interquartile range

Midhinge

Question 6:

A data in which we study about provinces is called.....

Quantitative

Qualitative

Chronological

Geographical ✓

Question 7:

First quartile for 10.1, 10.3, 10.5, 10.5, 10.7, 10.9 is approximately .....

10.1

10.3 ✓

10.5

10.2

Question 8:

.....use the division of a circle into different sectors.

Frequency Polygon

Line graph

Conversion Graphs

Sector graphs ✓

Question 9:

..... is the mean of inverse values.

Harmonic mean ✓

Geometric mean

Midhinge

Arithmetic mean

Question 10:

In Positively Skewed Distribution, which of the following is true

Mode > median > mean

Mean > median > mode ✓

Mean > mode > median

None of these

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A histogram is a special kind of .....

Line graph

Sector graph

Picture graph

Bar graph ✓

Geometric mean of 130, 140, 160 is -----

135.5

142.8 ☒

130.5

145.7

Under which of the following conditions for three terms a, b, c, their all central tendencies (mean, median & mode) are equal

$a < b < c$

$a = b = c$  ☒

$a > b > c$

$a > b$  but  $b < c$

A pictograph also called .....

Pictogram ☒

Column Graphs

Histograms

Conversion Graphs

In an ordered array of data having even number of values, median is the . . .

Sum of 2 middle values.

difference of 2 middle values.

average of 2 middle values. ☒

average of 4 middle values.

.....graphs are similar to bar graphs.

conversion

column ☒

line

sector

In stem and leaf display stem shows

Thousandth number

None of these

Decimal number

Unit number ☒

A diagram that uses rectangles to represent frequency is called ..... graph

column and Bar

picture

histogram ☒

Sector

The measure of central tendency which divides the data into two equal parts is called.....

Mean

Geometric mean

Mode

Median ☒

The graph below shows the rainfall over a period of 12 months. Find the month in which there was the second least rainfall.

November

August ☒

January

June

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A diagram that uses rectangles to represent frequency is called .... graph

☐ Sector

☒ histogram

☐ picture

☐ column and Bar

2. The value that occurs most frequently in the data is called ...

☒ Mode

☐ Median

☐ Range

☐ Mean

3. \_\_\_\_\_ use the division of a circle into different sectors.

☒ Sector graphs

☐ Frequency Polygon

☐ Line graph

☐ Conversion Graphs

4. The graph below shows noon temperatures for five days. What weekday was it when the noon temperature started to increase for the first time?

☐ Tuesday

☒ Sunday

☐ Wednesday

☐ Thursday

5. Which of the following is true?

☐ None of these.

☐  $A.M < G.M < H.M$

☒  $A.M > G.M > H.M$

☐  $A.M = G.M = H.M$

6. In stem and leaf display stem shows:

- ☐ None of these
- ☒ Unit number
- ☐ Thousandth number
- ☐ Decimal number

7. Last year a small statistical consulting company paid each of its five statistical clerks \$22,000, two statistical analysts \$50,000 each, and the senior statistician/owner \$270,000. The number of employees earning less than the mean salary is:

- ☒ 5
- ☐ 4
- ☐ 7
- ☐ 6

Question #8

Question:

is used to calculate

Options:

- None of above
- Geometric Mean
- Harmonic Mean
- Arithmetic Mean

Correct Option: Harmonic Mean

Question #9

Question:

A, B, C = 360 degrees

Options:

Parabola

Sphere

Circle

Cone

Correct Option: Circle

Question #10

Question:

In an ordered array of data having even number of values median is equal to:

Options:

Difference of 2 middle values

Average of 4 middle values

Average of 2 middle values

Sum of 2 middle values

Correct Option: Average of 2 middle values

mt302

Question 1: A data in which we study about religion is called ....

Options:

Geographical

Qualitative ☒

Quantitative

Chronological

Question 2: Another name of mean in statistics is

Options:

Mode

None of these

Median

Expected value ☒

Question 3: What are the central tendencies?

Options:

none of them

Mean, median, mode and midrange ☒

Mean, median, mode and range

Mean, median, mode and harmonic mean

Question 4: Last year a small statistical consulting company paid each of its five statistical clerks \$22,000, two statistical analysts \$50,000 each, and the senior statistician/owner \$270,000. The number of employees earning less than the mean salary is:

Options:

6

5 ☒

7

4

Question 5: A ..... is a graphical display of a frequency table.

Options:

frequency polygon

histogram ☒

Sector

Ogive

Question 6: Geographical data deals with...

Options:

Regions ☒

Religion

Height

Income

Question 7: A diagram that uses rectangles to represent frequency is called .... graph.

Options:

column and Bar

picture

Sector

histogram ☒

Question 8: The depth of a river at four different points is 2, 7, 5, 6 feet. How feet long people can cross it?

Options:

6

7

5

4.5 ☒

Question 9: The degree to which numerical data tend to spread about an average is called

Options:

The dispersion ☒

None of these

Standard deviation

Correlation

Question 10: Mode of a set of values can be ....

Options:

Three

Exactly one

Two

More than one ☒

MTH302

Here are all the MCQs with their options and the correct option indicated with a tick mark:

Question 1:

When we calculate the mode of the data then it is necessary that first we

- a) arrange the data in ascending order
- b) arrange the data in descending order
- c) a and b both are correct
- d) a and b both are wrong

Correct Option: d) a and b both are wrong

Question 2:

A graphical method of representing the sample points of a multiple-department is

- a) An ogive
- b) A histogram

c) A tree diagram

d) A frequency polygon

Correct Option: c) A tree diagram

Question 3:

In sector graphs a circle is divided into \_\_\_\_\_ sectors.

a) four

b) two

c) any number of

d) six

Correct Option: c) any number of

Question 4:

A linear scale always represents \_\_\_\_\_ graph.

a) sector

b) circle

c) line

d) conversion

Correct Option: d) conversion

Question 5:

In qualitative data we study about the

a) Weight

- b) Height
- c) Religion
- d) income

Correct Option: c) Religion

Question 6:

Under which of the following conditions for three terms a, b, c, all central tendencies (mean, median & mode) are equal

- a)  $a=b=c$
- b)  $a=b>c$
- c)  $a<b<c$
- d)  $a>b>c$

Correct Option: a)  $a=b=c$

Question 7:

The most common measure of central tendency, which is affected by extreme values is called \_\_\_\_\_.

- a) Median
- b) Range
- c) Mode
- d) Mean

Correct Option: d) Mean

Question 8:

Which of the following is NOT a measure of central tendency?

a) Standard deviation

b) Quartile

c) Mode

d) Percentile

Correct Option: a) Standard deviation

Question 9:

Degrees are used to present percentages in \_\_\_\_\_.

a) bar graph

b) none of these

c) line graph

d) sector graph

Correct Option: d) sector graph

Question 10:

Which of the following is true?

a) Arithmetic Mean is based on all and mode is on the middle values.

b) Arithmetic Mean is based on all and median on the middle value.

c) None of these

d) Median based on most frequent value and mode on middle value.

Correct Option: b) Arithmetic Mean is based on all and median on the middle value.

1. From the following MS-Excel data. If in the Exponential Smoothing then the Forecasting sale in D7 or week 3 is 4350655046503850.

Options:

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Correct Answer: None of these.

2. What is the value of  $nCn$  (combination) when  $n$  is equal to  $n$  ( $nCn$ )?

Options:

- A. 0
- B. 1
- C. None of these
- D.  $1+q$

Correct Answer: 1

3. If we have a set of 5 items and we want to find the number of permutations when choosing 3 items at a time, its Excel syntax will be .....

Options:

- A. =FACT(5)
- B. =PERMUT(5,3)
- C. =PERMUT(3,5)
- D. =COMB(5,3)

Correct Answer: =PERMUT(5,3)

4. The probability that on a given day there will be more than 8 people absent is:

Options:

- A. None of these.
- B. 0.1840
- C. 0.50
- D. 0.2709

Correct Answer: None of these.

5. The probability of "success"  $p$  is equals to  $1-q$ .

Options:

- A.  $1 - q$
- B. None of these
- C.  $1+q$
- D. 0

Correct Answer:  $1 - q$

6. Which of these numbers cannot be a probability?

Options:

- A. 0.5
- B. 1.001
- C. 0
- D. 1

Correct Answer: 1.001

7. How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

Options:

- A.  $10P5$
- B.  $5^{10}$
- C.  $10C5$
- D.  $10^5$

Correct Answer:  $10^5$

8. If A and B are mutually exclusive events then,

Options:

- A.  $P(A \text{ or } B) = P(A) - P(B)$
- B.  $P(A \text{ and } B) = P(A) + P(B) + P(A \text{ or } B)$
- C.  $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- D. None of these

Correct Answers:

- 1.  $P(A \text{ or } B) = P(A) + P(B)$
- 2.  $P(A \text{ and } B) = 0$

9. The number of telephone calls that pass through a switchboard has a Poisson distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:

Options:

- A. 0.1353
- B. 0.2707
- C. 0.0517
- D. 0.0183

Correct Answer: 0.1353

1. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:

Options:

- 0.1353
- 0.0183 (Correct)
- 0.0517
- 0.2707

2. What is the probability of  $r$  or more successes when  $m=7$ ;  $r=9$ ?

Options:

- 0.2709
- 0.1840
- 0.5

- None of these (Correct)

3. The set of collection of all possible outcomes of an experiment is called \_\_\_\_\_.

Options:

- Sample Space (Correct)
- Power set
- None of these
- Sample set

4. If 'n' is the total number of objects and 'P' for permutation, what is the formula for the number of 'n' permutations of 'n' different objects taken 'n' at a time?

Options:

- $(n-n)!$
- $n!$
- $(n-1)!$
- $0!$

5. A number from 1 to 11 is chosen at random. What is the probability of choosing an odd number?

Options:

- $5/11$  (Correct)
- $2/11$
- $6/11$
- $1/11$

6. When a coin is flipped repeatedly, what will be the probability of success in each flip that it lands on head?

Options:

- 0.511
- 0.2

- 0.5 (Correct)

7. State the addition theorem of probability for any two mutually exclusive events.

Options:

- $P(A \text{ or } B) = P(A) - P(B) + P(A \text{ and } B)$
- $P(A \text{ or } B) = P(A) - P(B)$
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$  (Correct)
- None of these

8. A number from 1 to 11 is chosen at random. What is the probability of choosing an even number?

Options:

- $1/11$
- $5/11$
- $2/11$  (Correct)
- $6/11$

9. Which of the following parameter/parameters are required for the "CRITBINOM" function?

Options:

- Number of Successes
- No. of Trials
- All of the above (Correct)
- Probability

10. A pair of fair dice is rolled. What will be the probability of the sum of dots on the upper sides being an odd number?

Options:

- $15/36$
- $18/36$  (Correct, simplifies to  $1/2$ )
- $10/36$

- None of these

1. How many words can be formed by using all letters of the word 'LAHORE'?

Options:

a) 120

b) 720

c) 520

d) 320

Correct Option: a) 120

2. In how many ways can the letters of the word 'LEADER' be arranged?

Options:

a) 360

b) 720

c) 72144

d) 44

Correct Option: b) 720

3. For the Poisson distribution,

Options:

a)  $E(X) = \mu$ ,  $\text{Var}(X) = -\mu$

b)  $E(X) = 0$ ,  $\text{Var}(X) = 0$

c)  $E(X) = -\mu$ ,  $\text{Var}(X) = \mu$

d)  $E(X) = \mu$ ,  $\text{Var}(X) = \mu$

Correct Option: d)  $E(X) = \mu$ ,  $\text{Var}(X) = \mu$

4. Five friends want to take lunch in a restaurant on a round table. So, in how many ways can they sit around the table?

Options:

- a) 5
- b) 12
- c) 20
- d) 24

Correct Option: d) 24

5. Probability distribution is a set of possible values that a random variable can assume and their associated probabilities of occurrence.

Options:

- a) True
- b) False

Correct Option: a) True

6. If we have a set of 5 items and we want to find the number of permutations when choosing 3 items at a time, its Excel syntax will be...

Options:

- a) =PERMUT(5,3)
- b) =COMB(5,3)
- c) =FACT(5)
- d) =PERMUT(3,5)

Correct Option: a) =PERMUT(5,3)

7. The Error in Exponential Smoothing can be calculated by the formula:

Options:

- a) Error = | Actual Sale - Forecast Sale |
- b) Error = | Actual Sale + Forecast Sale |
- c) Error = Actual Sale + Forecast Sale
- d) Error = Actual Sale - Forecast Sale

Correct Option: a) Error = | Actual Sale - Forecast Sale |

8. From the following MS-Excel data, if in Exponential Smoothing then the Forecasting sale in D7 or week 3 is...

Options:

- a) 4350
- b) 6550
- c) 4650
- d) 3850

Correct Option: This cannot be determined without the smoothing parameter ( $\alpha$ ).

9. If operation A can be performed in  $m$  ways and B in  $n$  ways, then the two operations can be performed together in \_\_\_\_ ways.

Options:

- a)  $m - n$
- b)  $m + n$
- c)  $m/n$
- d)  $m * n$

Correct Option: d)  $m * n$

10. What is the value of  ${}^5C_5$ ? (C for combination)

Options:

- a) 2
- b) 1
- c) 3
- d) 4

Correct Option: b) 1

\*\*MCQ 1:\*\*

Question: A worker out of 500 gets a prize by lottery. What is the chance of any one individual say Zain being selected?

Options:

- A.  $1/600$
- B.  $1/500$
- C.  $500/500$
- D.  $1/500$

Correct Answer: D

**\*\*MCQ 2:\*\***

Question: When two dice are rolled together, the outcomes on the faces are

Options:

- A. Mutually exclusive.
- B. True
- C. False
- D. Independent

Correct Answer: C

**\*\*MCQ 3:\*\***

Question: In the following dialog box, the output range must be on the \_\_\_\_ worksheet as the data used in the input range.

Options:

- A. Previous
- B. Next
- C. Same
- D. None of these

Correct Answer: C

**\*\*MCQ 4:\*\***

Question: For mutually exclusive events

Options:

- A.  $P(A \cup B) = P(A)P(B)$

B.  $P(A \cup B) = P(A) + P(B)$

C.  $P(A \cup B) = P(A)/P(B)$

D.  $P(A \cup B) = P(A) - P(B)$

Correct Answer: B

**\*\*MCQ 5:\*\***

Question: The formula  $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$  is called

Options:

A. The law of addition

B. None of these

C. The law of multiplication

D. The law of subtraction

Correct Answer: A

**\*\*MCQ 6:\*\***

Question: What is the range of probability of any event?

Options:

A. None of these

B. -1 to 0

C. 0 to 1

D. -1 to 1

Correct Answer: C

**\*\*MCQ 7:\*\***

Question: In Excel's NEGBINOMDIST function, what does the argument "number\_f" represent?

Options:

A. The cumulative probability

B. The threshold number of successes

C. The number of failures

D. The number of failures

Correct Answer: B

**\*\*MCQ 8:\*\***

Question: In how many ways can a cricket eleven be chosen out of a batch of seventeen players?

Options:

A.  ${}^{11}C_{11}$

B.  ${}^{17}C_4$

C.  ${}^{17}C_{11}$

D.  ${}^{11}C_{17}$

Correct Answer: C

**\*\*MCQ 9:\*\***

Question: What is the probability of  $r$  or more successes when  $m=7$ ;  $r=9$ .

Options:

A. 0.270

B. 0.184

C. 0.5

D. None of these

Correct Answer: D

**\*\*MCQ 10:\*\***

Question: When two dice are rolled together, the outcomes on the faces are

Options:

A. Mutually exclusive.

B. True

C. False

D. Independent

Correct Answer: C

1. The probability of getting at least 5 times head-on tossing an unbiased coin for 6 times by using the binomial distribution is

- A.  $P(X \geq 5)$
- B.  $P(X = 5) + P(X = 6)$
- C.  $P(X \geq 5)$  (Correct Option)
- D. All options are correct.

2. When two coins are tossed, then the probability that two heads are obtained is

- A.  $1/4$
- B.  $1/5$
- C.  $1/2$  (Correct Option)
- D.  $1/3$

3. An automatic production line breaks down every 2 hours. Special production requires uninterrupted operation for 8 hours. What is the probability that this can be achieved?

- A. 0.312
- B. 0.018 (Correct Option)
- C. None of these.
- D. 0.021

4. Which of the following is the best option for ' $\alpha$ ' in the formula "New forecast = old forecast +  $\alpha \times$  (old actual – old forecast)"?

- A. Smoothing constant (Correct Option)
- B. Mean square error
- C. Forecasting
- D. Exponential smoothing

5. The probability of successes can be less than 0 or greater than 1.

- A. False

- B. True (Correct Option)

6. In binomial distribution, each observation represents

- A. None of these
- B. Only one outcome ("success")
- C. Only one outcome ("failure")
- D. One of two outcomes ("success" or "failure") (Correct Option)

7. Faulty apple toffees in a production line average out at 6 per box. The management is willing to replace one box in a hundred. What is the number of faulty toffees that this probability corresponds to?

- A. 13
- B. 18 (Correct Option)
- C. None of these.
- D. 20

8. The set of collection of all possible outcomes of an experiment is called

- A. None of these
- B. Sample set
- C. Power set
- D. Sample Space (Correct Option)

9. When two dice are rolled together, the outcomes on the faces are

- A. Mutually exclusive. (Correct Option)
- B. True
- C. False

10. The permutation of three objects A, B & C are

- A. 9
- B. 6 (Correct Option)

- C. 8
- D. 8

1. How many words can be formed by rearranging the letters of the word ASCENT such that A and T occupy the first and last positions respectively?

Options:

- A) 4!
- B) 6!
- C) 5!
- D) 8!

Correct Option: A) 4!

2. The formula "`=BINOMDIST(3, 6, 0.5, TRUE)`" is for Cumulative Binomial distribution.

Options:

- A) False
- B) True

Correct Option: A) False

3. In binomial distribution, the experiment is repeated for a:

Options:

- A) Variable number of times
- B) Many number of times
- C) None of these
- D) Fixed number of times

Correct Option: D) Fixed number of times

4. Exponential smoothing method is applicable to the \_\_\_\_\_ series data.

Options:

- A) Time

- B) Geometric
- C) None of these
- D) Discrete

Correct Option: A) Time

5. The probability of successes can be less than 0 or greater than 1.

Options:

- A) True
- B) False

Correct Option: B) False

6. The reasonable value for the smoothing constant lies between:

Options:

- A) 0.2 to 0.3
- B) 2 to 3
- C) 1 to 2
- D) 0.1 to 0.2

Correct Option: D) 0.1 to 0.2

7. Which of the following parameter/parameters are required for the "CRITBINOM" function?

Options:

- A) Number of Successes
- B) Probability
- C) All of the above
- D) No. of Trials

Correct Option: C) All of the above

8. For the Poisson distribution,  $E(X) = \mu$ ,  $\text{Var}(X) = \mu$ .

Options:

A)  $E(X) = \mu$ ,  $\text{Var}(X) = \mu$

B)  $E(X) = -\mu$ ,  $\text{Var}(X) = \mu$

C)  $E(X) = 0$ ,  $\text{Var}(X) = 0$

D)  $E(X) = \mu$ ,  $\text{Var}(X) = -\mu$

Correct Option: A)  $E(X) = \mu$ ,  $\text{Var}(X) = \mu$

9. What is the value of  $nCn$ ? (C for combination)

Options:

A)  $n - 1$

B)  $n$

C) 0

D) 1

Correct Option: D) 1

10. The Excel command "`=BINOMDIST(-2, 5, 0.5, FALSE)`" gives an error of #NUM. The reason is that the last parameter is FALSE.

Options:

A) True

B) False

Correct Option: A) True

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